Digital Transformation — Module 06: Learning — Discussion Questions

20 questions on machine learning, model lifecycle, model drift, and MLOps.

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1. How does ML formalize the Active Inference learning process — updating generative models from data?
2. Zillow lost $381M because its pricing model drifted. How should organizations detect and prevent model drift?
3. Why is "garbage in, garbage out" the fundamental ML principle for organizations?
4. When should an organization build ML models in-house versus using third-party ML services?
5. How does the ML lifecycle (data → train → deploy → monitor → retrain) compare to traditional software development?
6. What organizational capabilities are needed before an ML initiative can succeed?
7. How do feedback loops in recommendation systems create self-reinforcing patterns?
8. When is simpler (logistic regression) better than complex (deep learning) for organizational ML applications?
9. How should organizations handle ML model failures — when the model produces wrong outputs?
10. What is "hidden technical debt" in ML systems and why is it particularly dangerous?
11. Assess your organization's ML maturity (ad hoc → repeatable → managed → optimized).
12. Identify a prediction task in your organization. What data would you need? What model approach?
13. Design a model monitoring system that detects drift before it causes significant business impact.
14. What ethical considerations arise from organizational ML deployment? How should they be governed?
15. Compare the training needs for data scientists versus business managers in an ML-enabled organization.

16. How should an organization decide when to retrain versus rebuild a model?
17. Design an MLOps pipeline for a specific organizational ML use case.
18. How do data labeling requirements affect the feasibility of supervised ML in your organization?
19. When has organizational ML created unintended consequences? Analyze a known case.
20. Design a "model governance" framework for an organization deploying 20+ ML models simultaneously.